TEST CASES

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Stock Management	To test stock deduction in the sales service after a sale	Product "Wireless Router" exists in DB with stock = 10	1. Call create_sale() with product_id and quantity = 4 2. Query updated product stock in DB	Product: Wireless Router Price: 69.99 Initial stock: 10 Qty sold: 4	Sale record is returned and product stock updates from 10 to 6	create_sale() completed the sale and product stock decreased from 10 to 6.	**PASS**
UT_02	Sales Validation	Reject sale exceeding available stock	Product "Mousepad" exists in DB with available stock = 4	1. Call create_sale() requesting 5 units when available stock is 4 2. Verify exception and inspect DB	Product: Mousepad Available stock: 4 Requested: 5	SaleValidationError ("Insufficient stock...") is raised; sale is rejected and stock remains 4	Overselling was rejected with SaleValidationError; no sale was created and stock remained 4.	**PASS**
IT_01	User Authentication	To test login functionality with valid credentials	Active user account (test-staff) exists in DB	1. Navigate to login page (/login) 2. Enter valid username and password 3. Submit login (POST /login)	Username: test-staff Password: StaffPass123	User is authenticated, session is established, and user is redirected to dashboard (/)	Manual: Redirected to dashboard (/). Automated: POST /login returned HTTP 200 and established session (dashboard content not asserted by test).	**PASS**
IT_02	User Authentication	To test login with invalid password	User account (deshna-test) exists in DB	1. Navigate to login page (/login) 2. Enter valid username and wrong password 3. Submit login (POST /login)	Username: deshna-test Password: WrongPassword123	Error message: "Invalid username or password, or account deactivated."	Login was rejected and "Invalid username or password, or account deactivated." was displayed.	**PASS**
IT_03	Product Management	To test adding a new product with valid data	Create Product form (/products/create) is accessible; SKU SPK-BT-100 is unused	1. Navigate to Create Product (/products/create) 2. Enter product details 3. Click Create Product	Name: Bluetooth Speaker SKU: SPK-BT-100 Price: 39.99 Qty: 20	Product is saved in DB and flash message "Product created successfully." is displayed	Product was saved in DB with SKU SPK-BT-100, price 39.99, stock 20, and confirmation message was displayed.	**PASS**
IT_04	Product Management	To test adding a product with duplicate SKU	Product "First Item" with SKU UNIQUE-SKU already exists in DB	1. Navigate to Create Product (/products/create) 2. Enter existing SKU UNIQUE-SKU 3. Click Create Product	Name: Second Item SKU: UNIQUE-SKU (duplicate) Price: 20.00 Qty: 3	Error message: "A product with this SKU already exists." and duplicate is not saved	Duplicate SKU was rejected with "A product with this SKU already exists."; no duplicate product was created.	**PASS**
IT_05	Input Validation	Verify clear validation for invalid product or sale input	Create Product (/products/create) and Create Sale (/sales) forms are accessible	1. Submit blank product name or zero sale quantity 2. Inspect response and DB records	Product name: " " (blank) Sale quantity: 0	Clear validation message is displayed ("Product name is required." / positive quantity); invalid record is not saved	Invalid input was rejected with a validation message and no invalid product or sale record was saved.	**PASS**
IT_06	Audit Logging	Verify product or stock changes create an audit entry	Authorized user (test-admin) is logged in; product "Audit Mouse" exists with stock = 5	1. Update product stock from 5 to 10 via POST /products/<id>/edit 2. Query AuditLog model/table in DB	Product: Audit Mouse Stock update: 5 to 10	AuditLog DB record stores action="stock_edit", actor user_id, timestamp, and details "5 to 10"	Stock update created an AuditLog record in DB with the authenticated actor user_id and "Stock changed from 5 to 10".	**PASS**
IT_07	Billing + Inventory Integration	To test invoice generation and stock synchronization together	Product "Keyboard" exists in DB with price = 25.50 and stock = 5	1. Submit sale for 2 units (POST /sales) 2. Follow redirect to invoice (/sales/<id>/invoice) 3. Verify invoice and DB stock	Product: Keyboard Price: 25.50 Initial stock: 5 Qty: 2	Invoice is generated ("Sales Invoice", subtotal 51.00) and stock is reduced by 2 (from 5 to 3) simultaneously	Invoice was generated and displayed, and inventory stock in DB was reduced from 5 to 3.	**PASS**
IT_08	Purchase Order + Inventory Integration	To test stock update when a Purchase Order is marked as Received	Manager or Admin logged in; Pending PO (PO-0001) exists for "Desk Mat" (stock = 14) with quantity 20	1. Open Purchase Orders 2. Mark PO-0001 as Received (POST /purchase-orders/<id>/receive)	PO ID: PO-0001 Product: Desk Mat Initial stock: 14 Qty: 20	PO status becomes "Received" and product stock increases by 20 (from 14 to 34)	Purchase Order PO-0001 receipt completed and the received quantity (20) increased stock from 14 to 34.	**PASS**
IT_09	Authentication + Role Access Integration	To test restricted access enforcement across modules	Staff account (test-staff) is logged in	1. Log in as Staff (test-staff) 2. Attempt to access Supplier module (GET /suppliers)	Username: test-staff Role: staff Route: /suppliers	Access denied with HTTP 403 Forbidden response.	Staff access to the restricted Supplier module (/suppliers) was denied (HTTP 403).	**PASS**

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_10	Sales Transaction	Verify rollback of a failed multi-item sale	Multiple products exist in DB (Keyboard stock = 5, Monitor stock = 0)	1. Submit a multi-item sale where one item exceeds available stock 2. Inspect Sale, SaleItem, and Product records	Item 1: Keyboard (stock 5, qty 2) Item 2: Monitor (stock 0, qty 1)	No partial update; stock remains unchanged (5 and 0) and no incomplete sale is committed	Failed multi-item sale was rolled back; stock and sale records were not partially updated.	**PASS**
IT_11	Returns / Refunds	Verify return processing updates stock and records refund	Completed sale exists for "USB-C Hub" (unit price 25.00, sold 3, stock = 7); return quantity is valid	1. Return 1 unit from the completed sale item 2. Inspect product stock and ProductReturn record	Product: USB-C Hub Unit price: 25.00 Return quantity: 1	Stock increases by 1 (from 7 to 8) and ProductReturn records quantity = 1 and refund_amount = 25.00	Valid return was processed; stock increased to 8 and ProductReturn recorded refund_amount 25.00.	**PASS**
IT_12	Database Backup	Verify successful database backup generation	SQLite database contains known test record ("test record"); BACKUP_DIR is configured	1. Run create_backup() in backup_database 2. Check generated backup file 3. Verify contents and PRAGMA integrity_check	Temporary SQLite DB with table backup_test containing ("test record",)	Backup file is created, contains the source record ("test record",), and passes SQLite integrity check ("ok")	Backup file was created; the known record was present and SQLite integrity check returned ok.	**PASS**
ST_01	Low Stock Alert System	To test alert generation when stock crosses the threshold	Product "Wireless Earphones" has reorder threshold = 8 and initial stock = 10	1. Reduce stock below 8 through a sale of 3 units (10 to 7) 2. Check dashboard (GET /) for low-stock alert	Product: Wireless Earphones Threshold: 8 Stock reduced from 10 to 7	Low-stock alert banner ("Low-Stock Alert") is displayed on the dashboard for "Wireless Earphones"	Low-stock alert was displayed on the dashboard when stock fell from 10 to 7 (below threshold 8).	**PASS**
ST_02	Reporting Module	To test sales report generation for a selected date range	Manager or Admin logged in; sales transaction ("USB Keyboard", 60.00 on 2026-10-12) exists in system	1. Navigate to Sales Report (/reports/sales) 2. Select date range 3. Click Generate Report	Date range: 2026-10-10 to 2026-10-15	Sales report is generated and displays matching transactions ("USB Keyboard") and calculated total (60.00) for the selected range	Sales report displayed the matching sale details ("USB Keyboard", 2026-10-12) and calculated total (60.00).	**PASS**
ST_03	Role Access	Verify Staff, Manager and Admin permissions	Test accounts for Staff (test-staff), Manager (test-manager), and Admin (test-admin) exist	1. Sign in using each role 2. Attempt permitted and restricted actions across modules	Staff, Manager, and Admin test accounts	Each role accesses permitted modules (HTTP 200) and is denied restricted actions (HTTP 403)	Role-based access checks passed; restricted actions were denied (HTTP 403) for unauthorized roles.	**PASS**
ST_04	Stock Reporting	Verify stock movement report calculations	Manager or Admin logged in; product "Report Mouse" (stock = 16) has sales, restocks, and returns in period	1. Open Stock Report (/reports/stock) 2. Select date range 2026-10-10 to 2026-10-15 3. Generate report and verify movement totals	Date range: 2026-10-10 to 2026-10-15 Sold: 2, Restocked: 5, Returned: 1 Current stock: 16	Report shows Units Sold (2), Restocked (5), Returned (1), Net Movement (4), and Current Stock (16)	Stock report displayed the movement categories (2, 5, 1), calculated net movement (4), and current stock (16) correctly.	**PASS**